

17-7 PH Stainless Steel, RH950, bar and forgings

Categories: [Metal](#); [Ferrous Metal](#); [Stainless Steel](#); [Precipitation Hardening Stainless](#); [T S10000 Series Stainless Steel](#)

Material Properties are measured in the longitudinal direction.

Notes:

Processing: RH950 - heated to austenitic range, 1040°C (1900°F), and water quenched. Reheated to 925°C (1700°F) to precipitate carbides. The austenite is transformed to martensite after cooling to room temperature and refrigerating at -75°C (-100°F). Tempered at 510°C (950°)

Applications: high strength high temperature applications, chemical processing equipment, heat exchangers, power boilers, superheater tubes

Corrosion Resistance: 17-7 PH is suitable for use in fresh water, industrial and marine atmospheres, and mild chemical and oxidizing environments. 17-7 PH should not be used in salt water or reducing environments.

Key Words: UNS S17700, double treatment alloy, RH950, bar and forgings, 17-7PH, 17-7 PH, 17/7PH, 17/7 PH, Precipitation Hardening

Vendors: [Click here](#) to view all available suppliers for this material.

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Physical Properties	Metric	English	Comments
Density	7.80 g/cc	0.282 lb/in ³	
Mechanical Properties	Metric	English	Comments
Hardness, Rockwell C	41	41	
Tensile Strength, Ultimate	1275 MPa	184900 psi	
Tensile Strength, Yield	1030 MPa	149000 psi	
Elongation at Break	6.0 %	6.0 %	
Reduction of Area	10 %	10 %	
Modulus of Elasticity	204 GPa	29600 ksi	
Fracture Toughness	76.0 MPa-m ^{1/2}	69.2 ksi-in ^{1/2}	Longitudinal, 44 HRC.
Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000830 ohm-cm	0.0000830 ohm-cm	
Thermal Properties	Metric	English	Comments
CTE, linear 	11.0 µm/m-°C @Temperature 0.000 - 100 °C	6.11 µin/in-°F @Temperature 32.0 - 212 °F	
	11.6 µm/m-°C @Temperature 0.000 - 315 °C	6.44 µin/in-°F @Temperature 32.0 - 599 °F	
Specific Heat Capacity	0.460 J/g-°C @Temperature 0.000 - 100 °C	0.110 BTU/lb-°F @Temperature 32.0 - 212 °F	
Thermal Conductivity 	16.4 W/m-K @Temperature 100 °C	114 BTU-in/hr-ft ² -°F @Temperature 212 °F	
	21.8 W/m-K @Temperature 500 °C	151 BTU-in/hr-ft ² -°F @Temperature 932 °F	
Melting Point	1400 - 1450 °C	2550 - 2640 °F	
Solidus	1400 °C	2550 °F	
Liquidus	1450 °C	2640 °F	
Component Elements Properties	Metric	English	Comments
Aluminum, Al	0.75 - 1.5 %	0.75 - 1.5 %	
Carbon, C	<= 0.090 %	<= 0.090 %	
Chromium, Cr	16 - 18 %	16 - 18 %	
Iron, Fe	70.6 - 76.8 %	70.6 - 76.8 %	as balance
Manganese, Mn	<= 1.0 %	<= 1.0 %	
Nickel, Ni	6.5 - 7.75 %	6.5 - 7.75 %	
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Silicon, Si	<= 1.0 %	<= 1.0 %	
Sulfur, S	<= 0.040 %	<= 0.040 %	

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.

